

3075. Maximize Happiness of Selected Children

Difficulty: Medium

Link: <https://leetcode.com/problems/maximize-happiness-of-selected-children/description/?envType=daily-question&envId=2025-12-25>

Generated at: 2025-12-25 15:26:58

Problem:

You are given an array `happiness` of length `n`, and a **positive** integer `k`.

There are `n` children standing in a queue, where the i^{th} child has **happiness value** `happiness[i]`. You want to select `k` children from these `n` children in `k` turns.

In each turn, when you select a child, the **happiness value** of all the children that have **not** been selected till now decreases by `1`. Note that the happiness value **cannot** become negative and gets decremented **only** if it is positive.

Return the **maximum** sum of the happiness values of the selected children you can achieve by selecting `k` children.

Example 1:

Input: `happiness = [1,2,3]`, `k = 2`

Output: 4

Explanation: We can pick 2 children in the following way:

- Pick the child with the happiness value == 3. The happiness value of the remaining children becomes [0,1].
- Pick the child with the happiness value == 1. The happiness value of the remaining child becomes [0]. Note

that the happiness value cannot become less than 0.

The sum of the happiness values of the selected children is $3 + 1 = 4$.

Example 2:

Input: happiness = [1,1,1,1], k = 2

Output: 1

Explanation: We can pick 2 children in the following way:

- Pick any child with the happiness value == 1. The happiness value of the remaining children becomes [0,0,0].
- Pick the child with the happiness value == 0. The happiness value of the remaining child becomes [0,0].

The sum of the happiness values of the selected children is $1 + 0 = 1$.

Example 3:

Input: happiness = [2,3,4,5], k = 1

Output: 5

Explanation: We can pick 1 child in the following way:

- Pick the child with the happiness value == 5. The happiness value of the remaining children becomes [1,2,3].

The sum of the happiness values of the selected children is 5.

Constraints:

- $1 \leq n == \text{happiness.length} \leq 2 * 10^5$
- $1 \leq \text{happiness}[i] \leq 10^8$
- $1 \leq k \leq n$